# ARJUN YADAPADITHAYA

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## EDUCATION

**Michigan Technological University | Houghton, MI** *2026–Present*
Ph.D. Student, Electrical and Computer Engineering | Research: Hardware-in-the-Loop co-design for smart-inverter integration and analog front-end validation.

**Illinois Institute of Technology | Chicago, IL** *May 2025*
M.S. Electrical and Computer Engineering, GPA: 3.9/4.0 | Relevant coursework: Advanced Computer Architecture and Hardware Security, Hardware-Software Co-Design, Digital Signal Processing, Computer Vision.

**NMAM Institute of Technology | Karnataka, India** *June 2021*
B.E. Electronics and Communication Engineering, GPA: 3.7/4.0 | Relevant coursework: VLSI Circuits, FPGA Design, Embedded Systems, PIC/ARM Microcontrollers.

## RESEARCH & ENGINEERING EXPERIENCE

**Doctoral Researcher | Michigan Technological University** *2026–Present*
- Designed and validated an analog front end interfacing a LEM current sensor with a TI TMS320F28379D C2000 DSP using passive level shifting, OPA328 buffering, and active anti-alias filtering.
- Working on experimental hardware validation and collaborating on a three-phase PV testbed integrating a PV emulator, inverter, and bidirectional DC–DC converters.
- Reviewed C-V2X cooperative-perception architectures, including Where2comm, When2com, V2X-ViT, SyncNet, CoAlign, PHCP, and F-Cooper.

**Research Engineer | Illinois Institute of Technology** *Jan 2024–Apr 2026*
- Designed, assembled, and brought up custom PCBs for a low-cost steam-trap monitoring system integrating thermocouples, piezoelectric accelerometers, and an ATmega-based microcontroller.
- Debugged SPI/$I^2C$ hardware-software integration using oscilloscopes and logic analyzers; developed sensor logging and monitoring workflows with InfluxDB and Grafana.
- Processed more than 160,000 logged sensor records to support monitoring and fault-analysis workflows.

**Senior Software Engineer | Capgemini | Bengaluru, India** *Aug 2021–Aug 2023*
- Automated XML-to-Excel reporting, reducing manual processing time by 25%.
- Improved backend performance and supported migration of deployment workflows from Jenkins to Rancher for scalable application deployment.
- Worked with cross-functional teams on backend data structures, application features, and system efficiency.

## PROJECTS

**Dynamic Branch Predictors | Verilog, Python, ModelSim** *Jan 2025–Apr 2025*
- Designed and verified gshare, gselect, and gpredict correlated dynamic branch predictors under a 4-bit global history register (GHR) and 16-entry branch history table (BHT).
- Analyzed indexing, aliasing, and history-depth trade-offs; built a reusable Verilog testbench and Python trace generator producing 6,000-line traces, then verified counter saturation and prediction timing in ModelSim.

**Battery-Operated Radiator Control | C/C++, ATmega, RFM69HCW** *Oct 2023–Apr 2026*
- Developed firmware for actuator control over a 270° range and implemented packet recovery for reliable 915 MHz-class sub-GHz RFM69HCW communication.
- Used watchdog-timer interrupts and deep-sleep cycles to reduce active duty cycle while maintaining periodic telemetry to InfluxDB.
- Conducted real-world validation in a high-traffic school environment; achieved a 74% energy-consumption reduction and won $30,000 at the 2024 Kaplan Pitch Tank.

**Smart Energy Meter | LM2902, ESP32, Wi-Fi, Grafana** *Mar 2024–Apr 2026*
- Built an analog signal-conditioning interface using an LM2902 op-amp and an ESP32-based Wi-Fi logger for energy data collection, analysis, and Grafana visualization.

## PUBLICATION

- Co-author, “Standardizing Grid Outage Resilience as a Building Performance Outcome: Phius REVIVE 2024,” accepted and published at CATE 2026.

## TECHNICAL SKILLS

**Programming/HDL**: C, C++, Python, MATLAB, Verilog, SQL | **Hardware & Design**: KiCad, EAGLE, LTspice, PCB design/assembly, schematic capture, BOM/component selection, oscilloscopes, logic analyzers, TI C2000/CCS | **Embedded & Interfaces**: ATmega, ESP32, ARM Cortex-M, FreeRTOS, SPI, $I^2C$, UART, PWM, Wi-Fi, Bluetooth, RFM69HCW | **Data & Software**: InfluxDB, Grafana, Flask, Linux, Git, MongoDB, Jenkins, Rancher, Angular, PyTorch